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Sanborn de Asis

What's the Latest on Palm OS® 5.x

A new name, Palm OS® Garnet

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What is Palm OS® Garnet, v.5.4?

Full platform release based on Palm OS® Garnet, v.5.3SC

- English, French, Italian, German, Spanish, Japanese, Chinese

Bug Fixes and Maintenance

Use current Palm OS 68K SDK, Release 3

Palm OS Garnet Simulator available

- Support 1x, 1.5x and 2x densities



Agenda

QVGA

PinMgr

Extended Text Manager Support

Fixes and Enhancements

Quarter VGA (QVGA)

1.5x density screen

- 240 x 320 including input area
- Active input with inking
- Maybe collapsible

Much like 2x density

- Bitmap and font families
- 160 x 160 standard coord grid

kb.palmsource.com article 495

- “Optimizing for QVGA Screens”



Quarter VGA (QVGA)

1.5x density screen

- 240 x 320 including input area
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Much like 2x density

- Bitmap and font families
- 160 x 240 in tall screen standard coords = different aspect ratio

kb.palmsource.com article 495

- “Optimizing for QVGA Screens”





QVGA Scaling

1.5 x odd number = rounding error

- Some things are rounded up while other things rounded down
- Coords are scaled then rounded up
- Rectangle Frame extent coords scaled then rounded down
- Window sizes are rounded up

QVGA Bitmap Scaling

Size of bitmap is rounded down

- 7 by 7 single density bitmap goes to 10 by 10 (not 11 by 11)

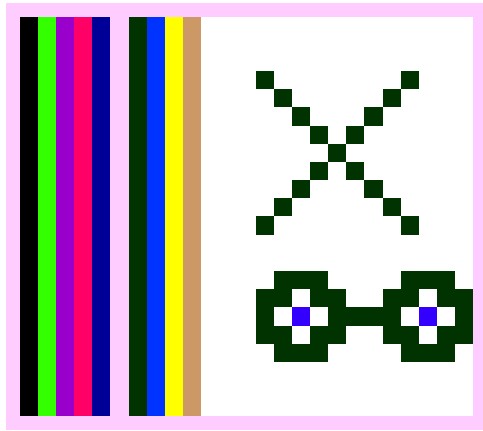
Every other pixel doubled for bitmaps

- Size clipped to fit “intended size”

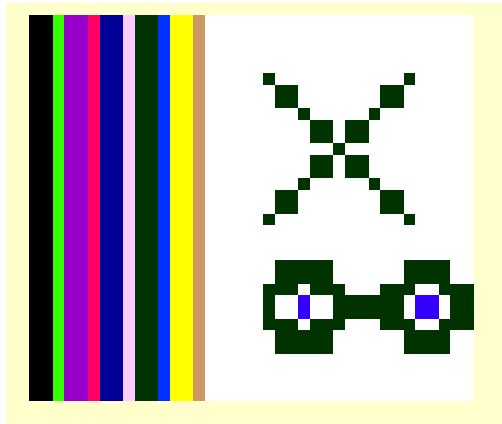
No 2x ↔ 1.5x bitmap scaling

Add 1.5X bitmaps to bitmap families

1X



1.5X



Standard Applications

PIM type applications mostly OK

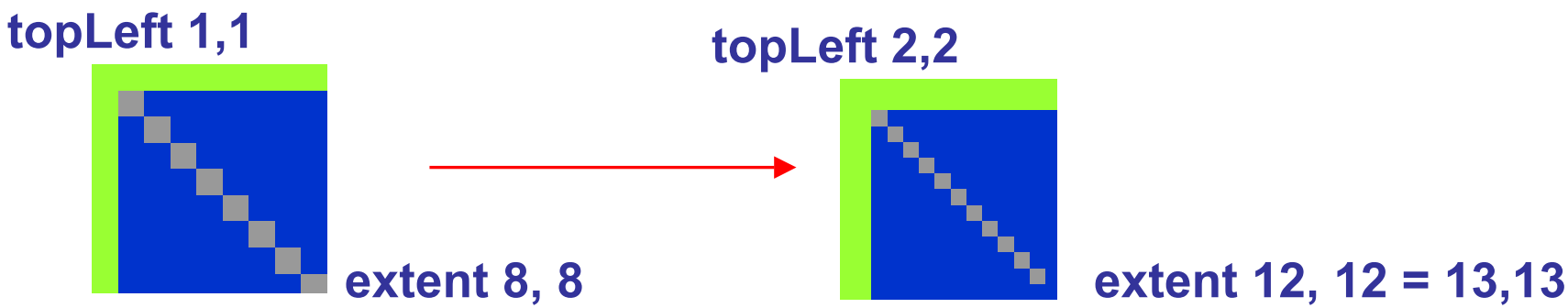
Some form objects might be shifted in respect to others

FormBitmaps need to have 1.5x members

Custom Fonts should also have 1.5x member

Custom Drawing may be off

- Consider using native coord system while drawing



Standard Applications

PIM type applications mostly OK

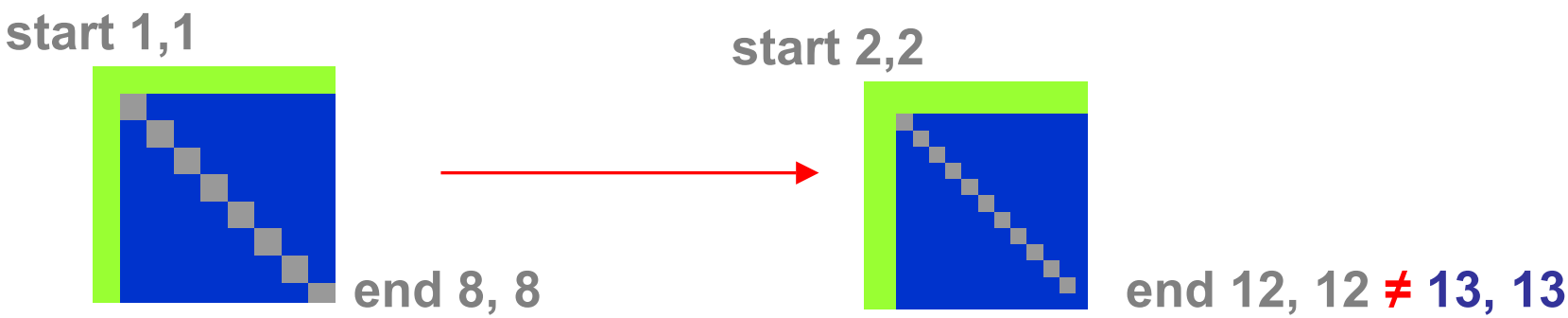
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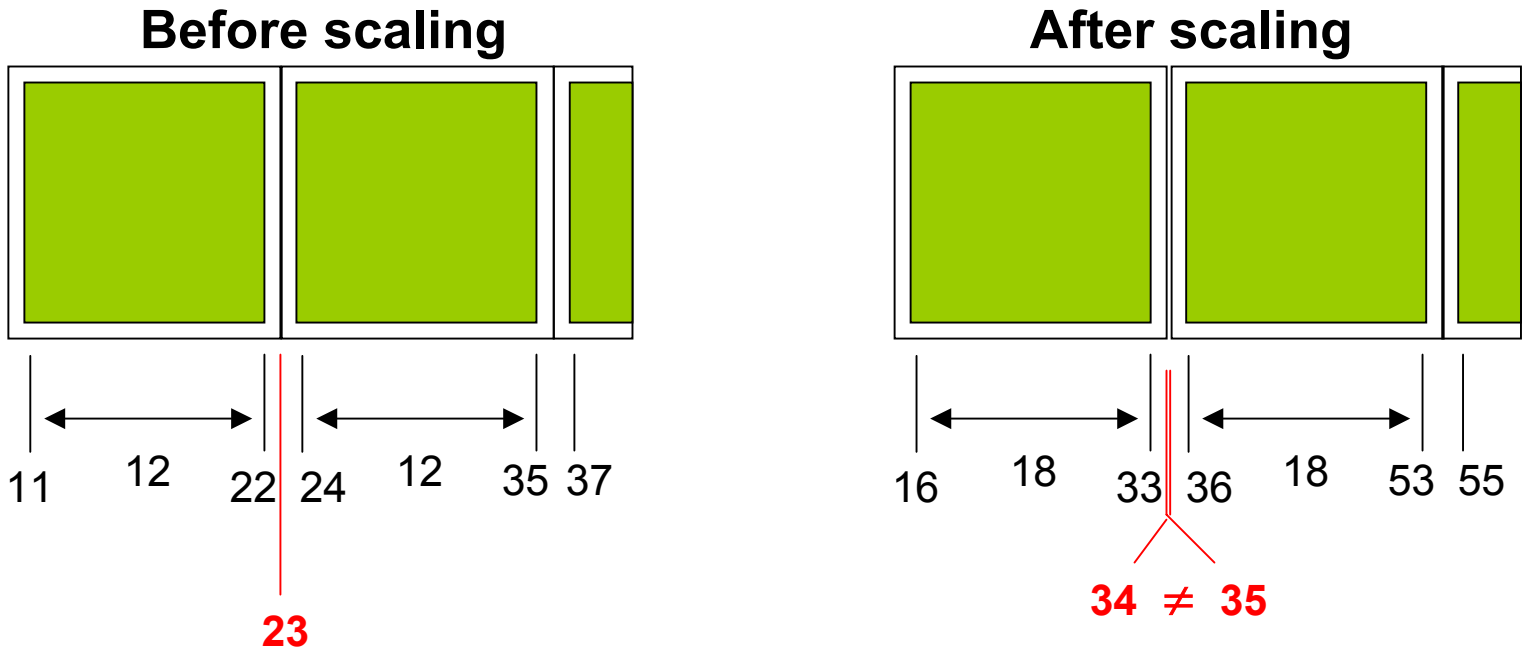
- Consider using native coord system while drawing



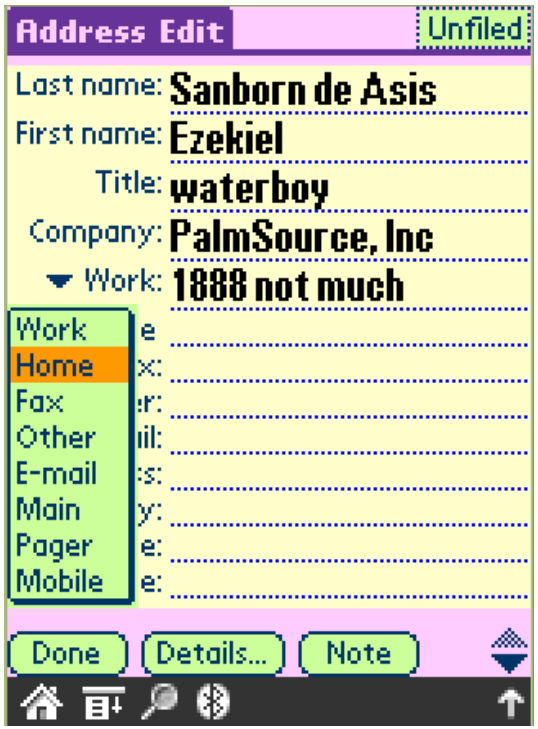
Push Button Example

Rectangle frames are drawn around UI objects bounds

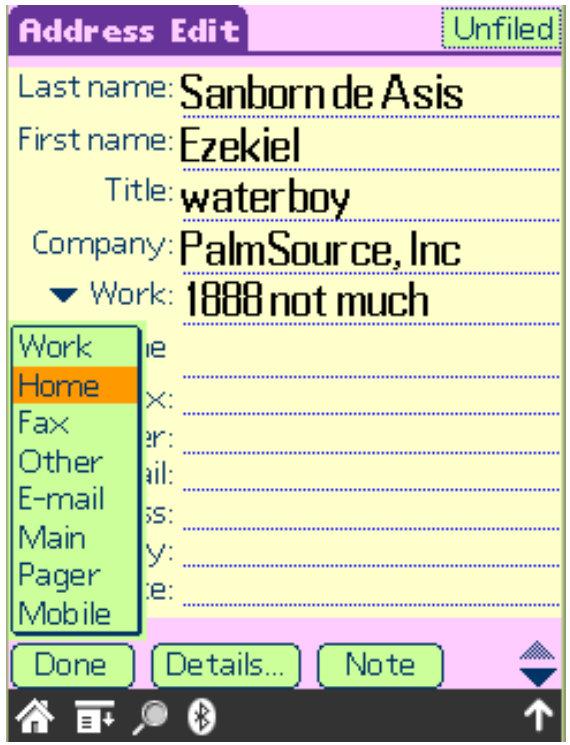
- Use odd extent values for push buttons to avoid this



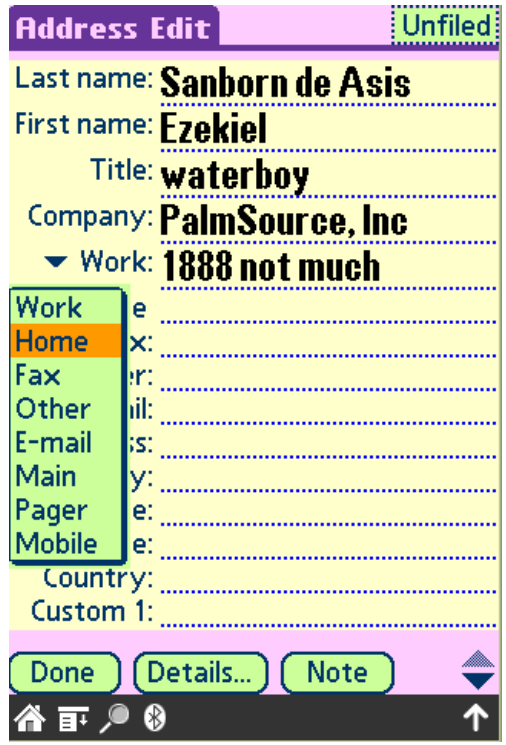
Address – with no modifications



1X



1.5X



2X



Offscreen windows and animation

Window format

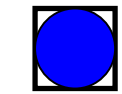
- screenFormat and genericFormat offscreens are 1x
- going from 1.5x to 1x and back to 1.5 will result in loss of data
- Use nativeFormat

Coordinate size of offscreen subject to coordinate scaling

- Consider WinSetCoordinateSystem to nativeCoordiantes before creating window to be exact
- Also can use nativeCoordiantes before copying back to screen
- Other option is to stick to even coordinates

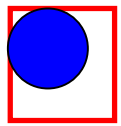
Complex Case

Avoid algorithms that rely on exact standard coordinate dimensions of objects

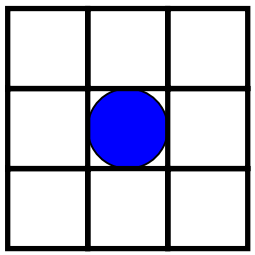


7 by 7

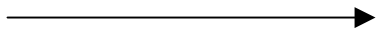
Ball
10 by 10

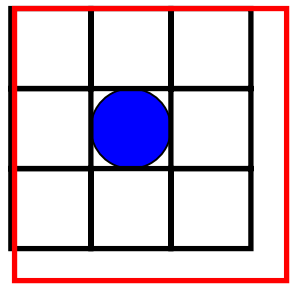


Window
11 by 11



21 by 21





Window
32 by 32



QVGA Quick Tips

Generally use even values for topLeft coords

Add 1.5x members to bitmap families

Be careful when form objects are close together

Use native coordinates for custom drawing

- Never leave OS in native coordinates

Use `nativeFormat` offscreen windows

Use `WinSetScalingMode()` to prevent bitmap scaling

Animation: move in even pixel amounts

Test on Palm OS Garnet Simulator

- Right click then Settings>Display>Resolution>240x320



Agenda

QVGA

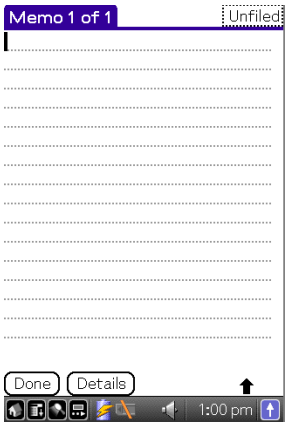
PinMgr

Extended Text Manager Support

Fixes and Enhancements

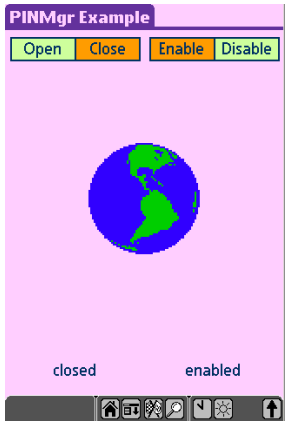
Pin Manager Evolution

Sony slk and vsk API



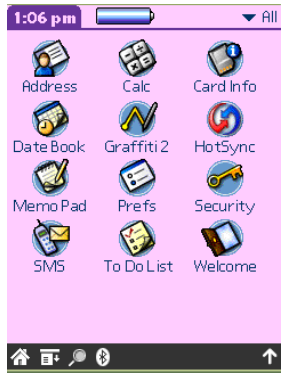
Sony

PInMgr 1.0

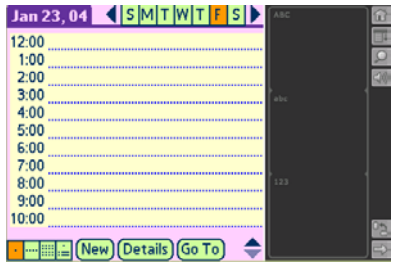


Garmin

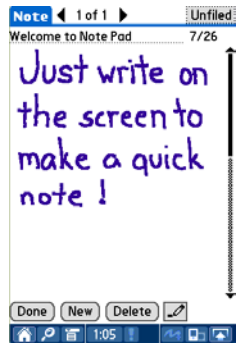
PinMgr 1.1



5.4



Tapwave



Tungsten T3

Dynamic Input Area (DIA)

2 New Features

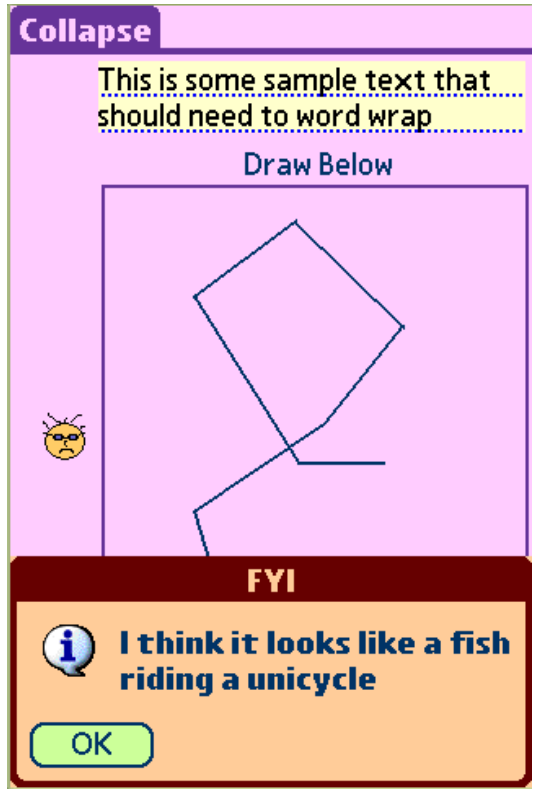
- Inking
- Collapsible

Existence of PinMgr doesn't mean the input area is collapsible

- Just that it is safe to use the calls

Article 507 in Knowledge Base

- "Using Palm OS PInMgr and cross platform DIA support"
- Adding basic support to an app can be very quick





PinMgr and Sony Vsk

Main differences

- For Sony API enable on app by app basis
- PinMgr enable form by form at frmOpenEvent
- Sony sends different notification at resize

Things in common

- Method to resize form and move objects

Sample code in Article 507 abstracts both APIs to `Collapse…` calls that work across different devices

- Sony, palmOne, Garmin, TapWave, Palm OS Cobalt
- Portrait and Landscape
- Most apps can just use these utility calls and work
- Other apps can modify utility code



Collapse - Set up and Clean up

At `AppStart` call `CollapseAppStart`

- Inits version of API
- Registers for notifications
- Opens Sony library if on Sony

At `AppStop` call `CollapseAppStop`

- Unregisters for notifications
- Restores state on older Sony API

At `frmOpenEvent` call `CollapseSetState`

- Boiler plate form set up for PinMgr API
- Sets initial input area state for form



Collapse - Reacting to a resize

At `frmOpenEvent` and `winDisplayChangedEvent` call `CollapseResizeForm()`

- Tries to resize form or dialog to currently available extent
- Returns true if resize needed and amount of horizontal and vertical resize offsets

If it returns true move form objects with `CollapseMoveResizeFormObject()`

- Object movement/resizing according to what makes sense for the form
- Use returned horizontal and vertical offsets as basis for movement and resizing
- Text fields: `FldRecalculateField()`

Post `winDisplayChangedEvent` in response to `winEnterEvent`

- In case display area changed outside of your event loop (alerts, etc.)



PinMgr Quick Tips

Remember, remember, remember … PinMgr calls operate on active form not DIA

Separate off form init call from for re-init

Avoid locking trigger for any length of time

Test in BOTH debug and release ROMs

Look at SampleCollapse code implementation

- Number of useful utility functions not mentioned here

Some UI Objects require special handling

- Fields, form bitmaps, scrollbars

Either always use, or never use, the userState



Agenda

QVGA

PinMgr

Extended Text Manager Support

Fixes and Enhancements



New and Improved Text Manager Routines

- **`TextXCharAttr`**
  - Returns GB related extended attributes
- **`TxtCharEncoding`**
  - Extended to include GB Character encoding
- **`TxtTransliterate`**
  - Also maps between single and double byte Latin characters
- **`TxtConvertEncoding`**
  - GB2312, GBK, ISO2022-CN, GSM to and from Unicode
  - Big5 (+HKSCS), ISO8859-1, US ASCII to Unicode



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Handwriting Recognition Extensibility

Notifications

- `SysNotifyGsiDrawIndicatorHandler`
- `SysNotifyInputAreaPendownEventHandler`
- `SysNotifyProcessPenStrokeEventHandler`
- `sysNotifyInputAreaDrawingEvent`
- `SysGetEventNotifyHandler`

Process Stroke PNO Thread (Not in Palm OS Cobalt)

- PceStartNativeThread



Enhancements

NetLib

- No Max limit on number of sockets.
- Dynamic NetLib stack size.
- Socket Notification

Graffiti® 2

- L – backspace – K fixed

UI

- PIMs support DIA
- Bluetooth Icon in Status Bar



Bug Fixes

Serial Wakeup handler crash

IP address settings byteswapped in 5.2 Netlib

Fatal alert shows when tap "discovery options" from bluetooth menu

DmSetDatabaseInfo causes system crash

SslWrite on a Non-blocking SSL Socket returns a -1 instead of the number of bytes written



Questions and Answers?

Come to the labs tonight for more info and help

Get the Palm OS® Garnet Simulator from the CD